

Strings in Java

Step 1: What is a String?

A String is a sequence of characters.

Example: "Hello", "Java"

Step 2: String is a Class

String is not a primitive data type. It is a class in Java.

Example:

```
String name = "Java";
```

Step 3: Ways to Create String

1. String s = "Hello";
2. String s = new String("Hello");

Step 4: String is Immutable

Strings cannot be changed once created.

Step 5: Input from User

```
import java.util.Scanner;
```

```
Scanner sc = new Scanner(System.in);
```

```
String name = sc.nextLine();
```

Step 6: Important Methods

1. length()

Returns length of string

Example:

```
String s = "Hello";  
System.out.println(s.length());
```

2. toUpperCase() / toLowerCase()

Example:

```
s.toUpperCase();
```

3. charAt()

Example:

```
s.charAt(1);
```

4. substring()

Example:

```
s.substring(1,4);
```

5. equals()

Used to compare strings

Example:

```
s1.equals(s2);
```

6. contains()

Example:

```
s.contains("Java");
```

7. replace()

Example:

```
s.replace("a","o");
```

8. trim()

Example:

```
s.trim();
```

Step 7: Looping through String

Normal Loop:

```
for(int i=0 ; i < s.length(); i++){  
    System.out.println(s.charAt(i));  
}
```

Enhanced Loop:

```
for(char ch : s.toCharArray()){  
    System.out.println(ch);  
}
```

Step 8: Programs

Program 1: Reverse String

```
String s = "Java";  
for(int i=s.length()-1;i>=0;i--){  
    System.out.print(s.charAt(i));  
}
```

Program 2: Palindrome

```
String s = "madam";  
String rev="";  
for(int i=s.length()-1;i>=0;i--){  
    rev += s.charAt(i);  
}  
if(s.equals(rev)){  
    System.out.println("Palindrome");  
}
```

Program 3: Count Vowels

```
String s="hello";
int count=0;
for(char ch : s.toCharArray()){
    if(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u'){
        count++;
    }
}
System.out.println(count);
```

Step 9: String vs StringBuilder

String → Immutable
StringBuilder → Mutable

Example:

```
StringBuilder sb = new StringBuilder("Hello");
sb.append(" Java");
```